

Large Degrees in Scale-Free Inhomogeneous Random Graphs

Joint work with Matthias Schulte (TUHH)

Chinmoy Bhattacharjee

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School of Mathematics and Statistics, University of Sheffield

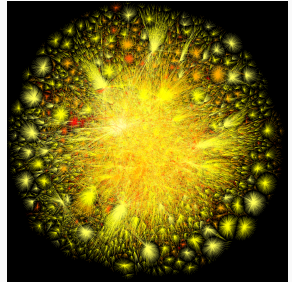
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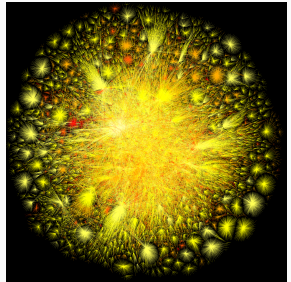


Internet 2010,
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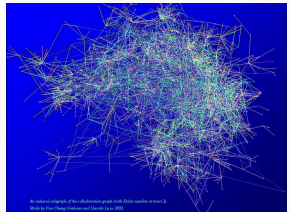
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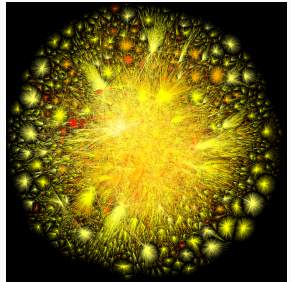
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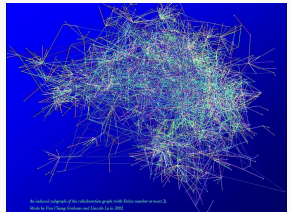
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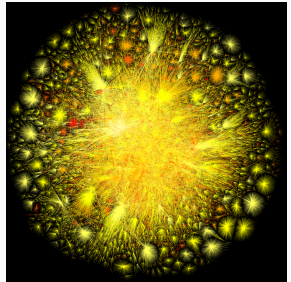
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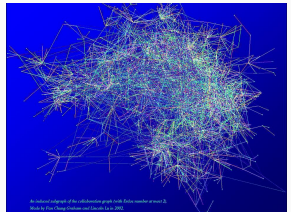
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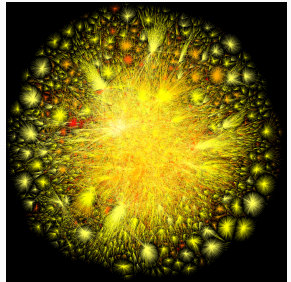
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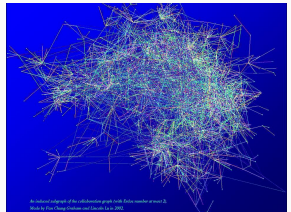
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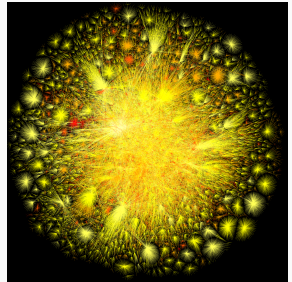
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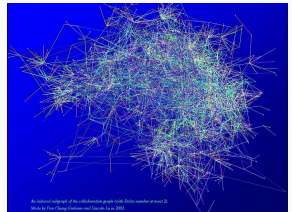
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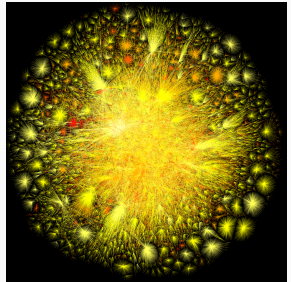
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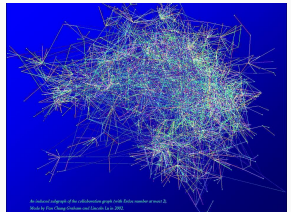
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Introduces inhomogeneity via **Hubs**.



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Scale-free random graph model

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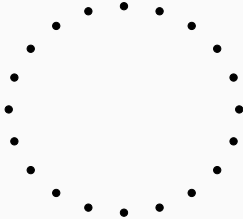
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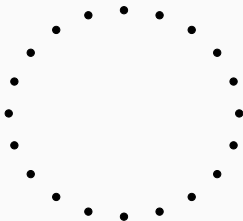
Possible to construct scale-free random graphs in a similar way as the Erdős-Rényi or random geometric graphs.

Erdős-Rényi graph



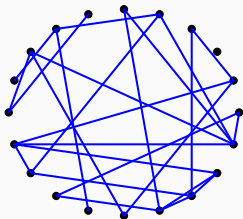
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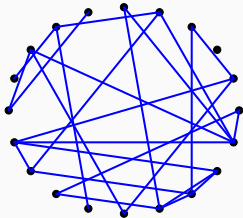
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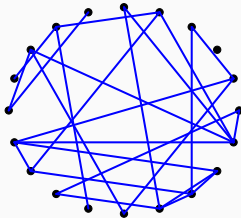
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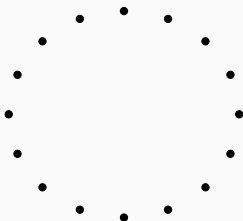
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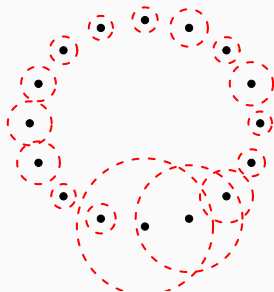


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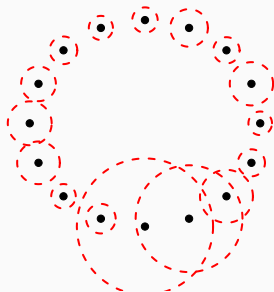
Associate i.i.d. weights having power law to each vertex. Moderates attractivity of vertices.



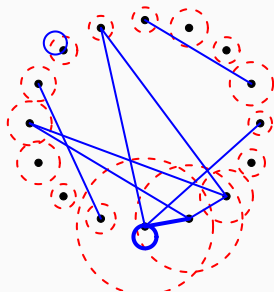
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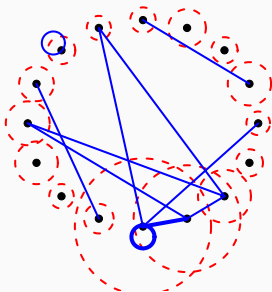
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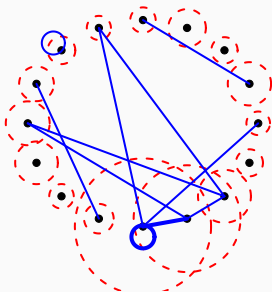


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Norros-Reittu model '06



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What about large degrees in NR model?

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(A1) Typical Degree and Weight: For each $n \in \mathbb{N}$, there exists (D_n, W) with $W \sim F_\beta$ such that for any $A \subseteq \mathbb{N}_0 \times [0, \infty)$ and $B \subseteq S$ with $\mathbb{E}|V_n \cap B| < \infty$,

$$\mathbb{E} \sum_{x \in V_n \cap B} \mathbb{1}\{(D_{n,x}, W_x) \in A\} = \mathbb{E}|V_n \cap B| \mathbb{P}\{(D_n, W) \in A\}.$$

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(A3) For each $m \in \mathbb{N}$, there exist non-negative constants a_m and C_m depending only on m such that

$$E_W |D_n - E_W D_n|^m \leq a_m W^{\frac{m\rho}{2}} + C_m \quad \mathbf{P}\text{-a.s.}$$

for all $n \in \mathbb{N}$.

Power-law degree distribution

Our assumptions are natural, and are satisfied not only by the Norros-Reittu model ('06), but also by

- Scale-free continuum percolation model on $\mathbb{R}^d$ (Deprez and Wüthrich '18),
- Scale-free percolation on $\mathbb{Z}^d$ (Deijfen, van der Hofstad and Hooghiemstra '13),
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Power-law degree (Bhattacharjee, Schulte '22)

Assume (A1)–(A3). Then for each $n \in \mathbb{N}$ the typical degree distribution follows a power-law with exponent $\gamma = \beta/p$, i.e.,

$$\mathbf{P}\{D_n > t\} \sim t^{-\gamma}, \quad \text{for } t > 0.$$

Main results: Poisson process limit for large degrees

Let $(S_n)_{n \in \mathbb{N}}$ be subsets of S such that $\mathbb{E}|\Delta_n| \approx n$ with $\Delta_n := V_n \cap S_n$.

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Poisson convergence of large degrees (Bhattacharjee, Schulte '22)

Let η_γ be a Poisson process with intensity measure ν_γ . Then under (A1)–(A3),

$$\bullet \left\{ \frac{D_{n,x}}{\xi q(n)}, x \in \Delta_n \right\} \cap (0, \infty) \xrightarrow{d} \eta_\gamma, \quad \text{as } n \rightarrow \infty.$$

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- $\xi^{-1} q(n)^{-1} \max_{x \in \Delta_n} D_{n,x} \xrightarrow{d} \text{Frechet}(\gamma), \quad \text{as } n \rightarrow \infty.$

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Let $D_{(1)}(n) \geq D_{(2)}(n) \geq \dots$ be the order statistics of $(D_{n,x})_{x \in \Delta_n}$.

Define the **Hill estimator** based on $k \in \mathbb{N}$ upper order statistics as

$$H_{k,n} = \frac{1}{k} \sum_{i=1}^k \log \frac{D_{(i)}(n)}{D_{(k+1)}(n)}.$$

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For preferential attachment model: Resnick and Wang '19.

Main idea

To prove the Poisson process convergence, we compare the point processes

$$\mathcal{D}_n := \left\{ \frac{D_{n,x}}{\xi q(n)} : x \in \Delta_n \right\} \quad \text{and} \quad \mathcal{E}_n := \left\{ \frac{W_x^p}{q(n)} : x \in \Delta_n \right\}$$

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In particular, we prove that, for any $k \in \mathbb{N}$,

$$\lim_{n \rightarrow \infty} \mathbf{P} \{ \text{node with } k\text{-th largest weight has } k\text{-th largest degree} \} = 1.$$

Thank you!

Reference: Bhattacharjee, C. & Schulte, M. (2022) Large degrees in scale-free inhomogeneous random graphs. *Ann. Appl. Probab.* **32**, 696–720.